

PROBABILITY AND STATISTICS

Lecture 6

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Outline I

- 1 Statistical Inference
- 2 Confidence Intervals (CI)
- 3 Hypothesis testing
- 4 Student's t test
- 5 Problems

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- 1 Statistical Inference
- 2 Confidence Intervals (CI)
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Why Statistical Inference?

Economists rarely observe entire populations.

Examples:

- Average wage in Ukraine.
- Inflation expectations of households.
- Monthly spending of consumers.
- Average return of an investment fund.

Instead, we collect a sample and draw conclusions about the population.

Sample $\Rightarrow$ Inference about Population

Population vs Sample

Population

- All households
- All firms
- All employees
- All investors

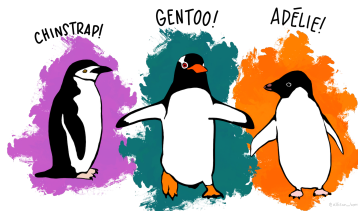
Sample

- 500 households
- 100 firms
- 200 employees
- 50 investors

Population parameters are usually unknown. μ , σ^2 , p .
We estimate them using sample statistics.

Parameter	Population	Sample Analogue	R Function
Mean	μ	$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$	<code>mean(x)</code>
Variance	σ^2	$s^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$	<code>var(x)</code>
St. D.	σ	$s = \sqrt{s^2}$	<code>sd(x)</code>
Proportion	p	$\hat{p} = \frac{x}{n}$	<code>mean(x == "succ</code>

Point Estimation



```
library(palmerpenguins)
library(ggplot2)
```

```
data("penguins")
head(penguins)
summary(penguins)
data <- na.omit(penguins)
```

```
mean(data$body_mass_g)
var(data$body_mass_g)
sd(data$body_mass_g)
mean(data$species=="Adelie")
```

```
> mean(data$body_mass_g)
[1] 4207.057
> var(data$body_mass_g)
[1] 648372.5
> sd(data$body_mass_g)
[1] 805.2158
> mean(data$species=="Adelie")
[1] 0.4384384
> |
```

How accurate is this estimate?

This motivates confidence intervals.

Outline

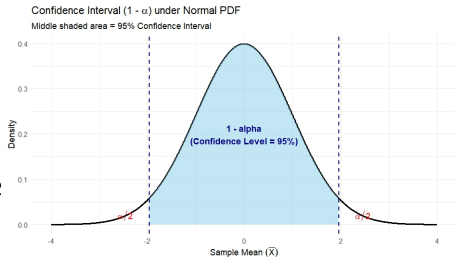
- 1 Statistical Inference
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CI for Sample Mean with known variance, Normal distribution

By CLT with known σ^2 :

$$\bar{X}_n \xrightarrow{d} \mathcal{N}\left(\mu, \frac{\sigma^2}{n}\right)$$

$$\begin{aligned} 1 - \alpha &= \\ &= P(-z_{1-\alpha/2} < Z < z_{1-\alpha/2}) \approx \\ &\approx P(-z_{1-\alpha/2} < \frac{\bar{X}_n - \mu}{\frac{\sigma}{\sqrt{n}}} < z_{1-\alpha/2}) \end{aligned}$$



$$= P\left(\bar{X}_n - z_{1-\alpha/2} \cdot \frac{\sigma}{\sqrt{n}} < \mu < \bar{X}_n + z_{1-\alpha/2} \cdot \frac{\sigma}{\sqrt{n}}\right),$$

where z_p is the p -th quantile of the standard normal distribution, with $p \in (0, 1)$: $\Phi(z_p) = p$.

Interpretation of a Confidence Interval

Incorrect interpretation:

There is a $1 - \alpha$ probability that the true mean lies inside the interval.

Correct interpretation:

If we repeatedly took samples and constructed confidence intervals, approximately $1 - \alpha$ of them would contain the true parameter.

<https://huggingface.co/spaces/aranaur/lotps-explorer>

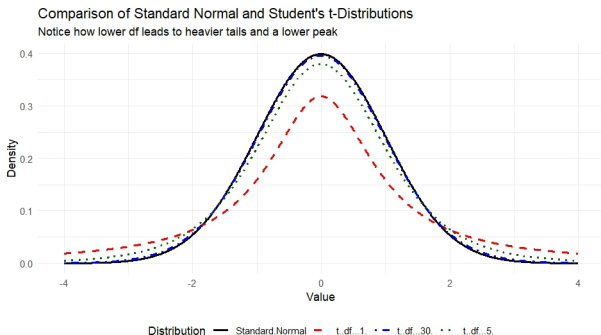
CI for Sample Mean with unknown variance, t- distribution

Consider a sample with normal distribution with unknown mean and variance. We should, first estimate it variance

$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2.$$

Then

$t_{n-1} = \frac{\bar{X} - \mu}{s} \cdot \sqrt{n}$ has t-distribution with n-1 degree of freedom (d.f.).



CI for Normal Sample with unknown variance

Then the interval(CI) with confidence level $1 - \alpha$ for the mean μ is

$$\bar{X}_n - t_{1-\alpha/2, n-1} \cdot \frac{s}{\sqrt{n}} < \mu < \bar{X}_n + t_{1-\alpha/2, n-1} \frac{s}{\sqrt{n}},$$

where n is a sample size, $t_{n-1, p}$ is a p -th quantile of t distribution with $n - 1$ degree of freedom (df).

Example

Let's construct 97% CI for body mass in Penguins dataset.

```
> n<-length(data$body_mass_g)
> alpha<-1-0.97
> ci<-c(mean(data$body_mass_g)-sd(data$body_mass_g)*qt(1-alpha/2, n-1)/sqrt(n),
+       mean(data$body_mass_g)+sd(data$body_mass_g)*qt(1-alpha/2, n-1)/sqrt(n))
> ci
[1] 4110.887 4303.227

> t<-t.test(data$body_mass_g, conf.level = 0.97)
> t$conf.int
[1] 4110.887 4303.227
attr(,"conf.level")
[1] 0.97
```

Confidence Intervals

Point estimate:

$$\hat{\mu} = 4207.057$$

does not quantify uncertainty.

A confidence interval provides a plausible range:

$$\text{Estimate} \pm \text{Margin of Error}$$

General form:

$$\text{Estimator} \pm \text{Critical Value} \times \text{Standard Error}$$

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Introduction to Hypothesis Testing

Hypothesis testing helps us evaluate claims.

Examples:

- Is inflation above 10%?
- Has unemployment changed?
- Is average income different from last year?
- Does a new policy affect spending?

Null and Alternative Hypotheses

Null Hypothesis H_0 represents the status quo.

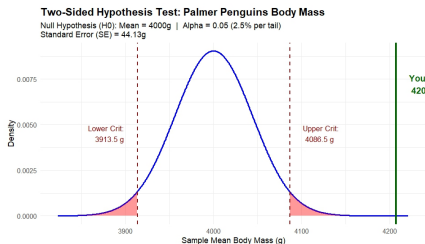
Alternative Hypothesis H_a represents evidence against the status quo.

Example for Penguins body mass:

$$H_0 : \mu = 4000$$

$$H_a : \mu \neq 4000$$

(As alternative we can also consider $\mu > 4000$ or $\mu < 4000$)



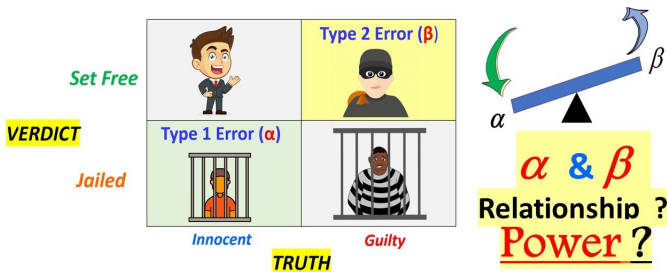
Courtroom Analogy

Defendant is presumed innocent

In statistics H_0 is assumed true until sufficient evidence suggests otherwise.

The burden of proof is on the data.

Type 1 Error & Type 2 Error



Economic Example of Type I Error

Government introduces a tax reform.

Hypotheses:

H_0 : No effect on economic growth

H_a : Positive effect

Type I Error:

Concluding the reform works when it actually has no effect.

Potential consequence:

- Incorrect policy decisions.
- Misallocation of resources.



Power of a Test

Power measures the ability to detect a real effect.

$$\text{Power} = 1 - \beta$$

High power means:

- Higher probability of detecting true effects.
- Better statistical decisions.

Power increases when:

- Sample size increases.
- Variability decreases.
- Effect size increases.

Significance Level

The significance level is α maximum acceptable probability of a Type I Error.

Common choices:

$$\alpha = 0.05, \quad \alpha = 0.01$$

Economics journals often use 5% as the standard.

P-Values

The p-value measures how surprising the observed data are under H_0 . So, **p-value** is a probability of obtaining your observed data (or something more extreme) assuming that the "null hypothesis" (the assumption that nothing new is happening) is true.

<https://www.youtube.com/watch?v=YMj79TfYUps>

Decision rule:

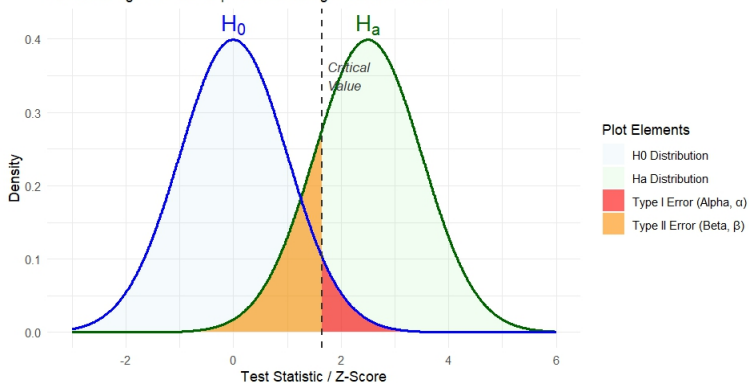
If $p < \alpha$ then reject H_0 .

Otherwise: If $p \geq \alpha$ then do not reject H_0 .

<https://huggingface.co/spaces/aranaur/lotps-explorer>

Hypothesis Testing: Comparing Distributions & Errors

Faint shading = Full Curve | Vibrant shading = Decision Errors



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Historical Background of the t-Test

Origin of the t-Test

The t-test was developed by **William Sealy Gosset** (1876–1937), a chemist and statistician working for the *Guinness Brewery* in Dublin, Ireland.

- At the beginning of the 20th century, industrial experiments often involved **small sample sizes**.
- Existing methods based on the normal distribution required the population variance to be known, which was rarely the case in practice.
- Gosset derived the distribution of

$$T = \frac{\bar{X} - \mu}{S/\sqrt{n}},$$

when the variance is estimated from the sample.

- Because Guinness prohibited employees from publishing research under their own names, Gosset published his work in 1908 under the pseudonym **“Student”**.

Sampling Distribution of the Mean

Impact

Student's t-test became one of the most widely used methods in statistics for comparing means when the population variance is unknown.

Let $X_1, \dots, X_n$ be i.i.d. observations with $E(X_i) = \mu$, $Var(X_i) = \sigma^2$.
The sample mean is

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i.$$

If $X_i \sim N(\mu, \sigma^2)$, then

$$\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right).$$

Why Student's t Distribution Appears

When σ is unknown we estimate it using

$$S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2.$$

The pivotal quantity becomes

$$T = \frac{\bar{X} - \mu}{S/\sqrt{n}}.$$

For normal data,

$$T \sim t_{n-1}.$$

The heavier tails account for the uncertainty in estimating σ .

As $n \rightarrow \infty$, $t_{n-1} \Rightarrow N(0, 1)$.

One-Sample t-Test

Hypotheses:

$$H_0 : \mu = \mu_0$$

vs

$$H_A : \mu \neq \mu_0 \quad (\mu > \mu_0 \text{ (greater)} \quad \text{or} \quad \mu < \mu_0 \text{ (less)})$$

Test statistic:

$$T = \frac{\bar{X} - \mu_0}{S/\sqrt{n}}.$$

Under H_0 :

$$T \sim t_{n-1}.$$

Reject H_0 if

$$|T| > t_{1-\alpha/2, n-1} \quad (T > t_{1-\alpha, n-1} \quad \text{or} \quad T < -t_{1-\alpha, n-1})$$

One-Sample Test: Actual Computation

$$H_0 : \mu = 4000$$

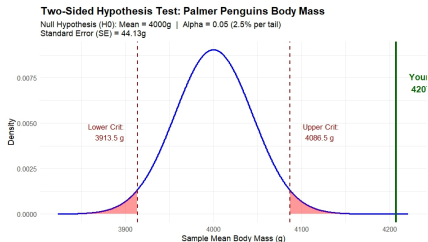
$$H_A : \mu \neq 4000.$$

```
t <- t.test(data$body_mass_g,  
            mu = 4000)  
t  
t$statistic  
t$p.value  
t$conf.int  
t$estimate
```

```
> t
```

One Sample t-test

```
data: data$body_mass_g  
t = 4.6925, df = 332, p-value = 3.952e-06  
alternative hypothesis: true mean is not equal to 4000  
95 percent confidence interval:  
 4120.256 4293.858  
sample estimates:  
mean of x  
 4207.057
```



Since $p\text{-value} < 0.05$, we reject H_0 . There is statistically significant evidence that the mean penguin body mass differs from 4000 g.

Exercise

Test the hypothesis that the population mean body mass of penguins is equal to 4200 g.

Question:

Do male and female penguins have equal mean body mass?

$$H_0 : \mu_{male} = \mu_{female}$$

vs

$$H_a : \mu_{male} > \mu_{female}$$

Independent Two-Sample Test

Suppose

$$X_1, \dots, X_{n_1}$$

and

$$Y_1, \dots, Y_{n_2}$$

are independent samples.

Hypotheses:

$$H_0 : \mu_X = \mu_Y.$$

Equivalent form:

$$H_0 : \mu_X - \mu_Y = 0.$$

Pooled Variance t-Test

Assume

$$\sigma_X^2 = \sigma_Y^2 = \sigma^2.$$

Pooled variance:

$$S_p^2 = \frac{(n_1 - 1)S_X^2 + (n_2 - 1)S_Y^2}{n_1 + n_2 - 2}.$$

Statistic:

$$T = \frac{\bar{X} - \bar{Y}}{S_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}.$$

Under H_0

$$T \sim t_{n_1+n_2-2}.$$

Welch Two-Sample Test

No equal variance assumption.

Statistic:

$$T = \frac{\bar{X} - \bar{Y}}{\sqrt{S_X^2/n_1 + S_Y^2/n_2}}.$$

Degrees of freedom:

$$\nu = \frac{(S_X^2/n_1 + S_Y^2/n_2)^2}{\frac{(S_X^2/n_1)^2}{n_1-1} + \frac{(S_Y^2/n_2)^2}{n_2-1}}.$$

(Satterthwaite approximation)

Two samples Student T-test in Practice

```
1 male_mass <- data$body_mass_g[
2   data$sex == "male"]
3
4 female_mass <- data$body_mass_g[
5   data$sex == "female"]
6
7 t.test(
8   male_mass,
9   female_mass,
10  alternative = "greater",
11  var.equal = TRUE)
```

Output:

Two Sample t-test

data: *male_mass* and *female_mass*

t = 8.5417, df = 331, p-value =
2.449e-16

alternative hypothesis: true difference in
means is greater than 0

95 percent confidence interval:
551.4399 Inf

sample estimates:

mean of x mean of y
4545.685 3862.273

Suppose output reports

$$t = 8.54, \quad p < 0.001.$$

Then:

- Strong evidence against H_0 .
- Male body mass is greater than and female body mass.
- The result is statistically significant.
- Confidence interval estimates the magnitude of the difference.

Paired t-Test

Paired observations:

$$(X_i, Y_i), \quad i = 1, \dots, n.$$

Define differences

$$D_i = Y_i - X_i.$$

Then test

$$H_0 : \mu_D = 0.$$

The statistic becomes

$$T = \frac{\bar{D}}{S_D / \sqrt{n}}.$$

Under normality:

$$T \sim t_{n-1}.$$

Comparison of t-Tests

Test	Samples	Parameter	DF
One-sample	1	μ	$n - 1$
Two-sample	2 independent	$\mu_1 - \mu_2$	$n_1 + n_2 - 2$
Welch	2 independent	$\mu_1 - \mu_2$	Satterthwaite
Paired	matched pairs	μ_D	$n - 1$

Robustness of the t-Test

The t-test is often robust to moderate violations of normality.

Reasons:

- By the Central Limit Theorem,

$$\bar{X}$$

approaches normality as n increases.

- Symmetric distributions rarely cause problems.
- Moderate skewness is usually acceptable for large samples.

Potential difficulties:

- Strong skewness.
- Heavy tails.
- Extreme outliers.
- Very small samples.

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Problem 1

The sleep Dataset

The built-in dataset `sleep` contains results from an experiment investigating the effect of two sleep-inducing drugs on 10 patients.

Variables:

- `extra`: increase in hours of sleep compared to baseline
- `group`: drug type (1 or 2)
- `ID`: patient identifier

Since each patient received both drugs, the observations are naturally paired.

Problem 1 -Statement

Task 1. Explore the dataset

- 1 Load the dataset.
- 2 Display all observations.
- 3 Examine its structure.

Task 2. One-Sample t-Test

Using Drug 1 observations, test whether Drug 1 increases sleep by more than 0 hours.

$$H_0 : \mu = 0$$

$$H_A : \mu > 0$$

Perform the test:

- manually;
- using `t.test()`.

Problem 1 -Statement (continued)

Task 3. Paired t-Test Determine whether there is a significant difference in effectiveness between Drug 1 and Drug 2.

Define

$$D = \text{Drug 2} - \text{Drug 1}.$$

Test

$$H_0 : \mu_D = 0$$

against

$$H_A : \mu_D \neq 0.$$

Construct a 95% confidence interval and interpret the results.

Problem 2

The ToothGrowth Dataset

The built-in dataset `ToothGrowth` investigates the effect of Vitamin C on tooth growth in guinea pigs.

Variables:

- `len`: tooth length
- `supp`: supplement type
 - OJ = Orange Juice
 - VC = Ascorbic Acid
- `dose`: dose level (0.5, 1, 2 mg/day)

Research question: **Does the delivery method (OJ vs VC) affect tooth growth?**

Problem 2- Statement

Task 1. Explore the dataset

- 1 Load the dataset.
- 2 Display all observations.
- 3 Examine its structure.

Task 2. Visualize the Data

Using *boxplot* inspect the differences between the groups

Task 3. Independent Two-Sample t-Test

- Compare the mean tooth lengths of the two independent groups using `t.test()` with significance level 0.1.

Hypotheses:

$$H_0 : \mu_{OJ} = \mu_{VC}$$

$$H_A : \mu_{OJ} \neq \mu_{VC}$$

- Construct a 90% confidence interval and interpret the results.

Problem 3 (optional)

A/B Testing

A/B testing is commonly used in business, marketing and web development. Users are randomly assigned to one of two webpage designs:

- **Control group:** users see the old webpage.
- **Treatment group:** users see the new webpage.

The goal is to determine whether the new webpage changes the conversion rate (e.g. made a purchase or signed up).

The Dataset

Dataset:

- Source: Kaggle E-commerce A/B Testing dataset
- Each row corresponds to one user.

Important variables:

- group: control or treatment
- landing_page: old or new page
- converted:
 - 1 = conversion occurred
 - 0 = no conversion

Problem Statement

Investigate whether the conversion rates differ between the two webpages.

Hypotheses:

$$H_0 : p_{\text{control}} = p_{\text{treatment}}$$

$$H_A : p_{\text{control}} \neq p_{\text{treatment}}$$

where

p

denotes the probability of conversion. Use significance level

$$\alpha = 0.05.$$

Problem 3-Statement

Task 1: Load the Data

[https://raw.githubusercontent.com/ksatola/](https://raw.githubusercontent.com/ksatola/Analyze-A-B-Test-Results/refs/heads/master/ab_data.csv)

[Analyze-A-B-Test-Results/refs/heads/master/ab_data.csv](https://raw.githubusercontent.com/ksatola/Analyze-A-B-Test-Results/refs/heads/master/ab_data.csv)

```
library(dplyr)
url <- "https://raw.githubusercontent.com/ksatola/Analyze-A-B-Test-Results/master/ab_
      data.csv"
ab_data <- read.csv(url)

# Display the first rows:

head(ab_data)
```

Task 2: Clean the Data

Keep only observations where

- control users saw the old page;
- treatment users saw the new page.

```
ab_clean <- ab_data %>%  
  filter( (group=="control" & landing_page=="old_page") | (group=="treatment" & landing  
    _page=="new_page") )
```

Task 3: Explore the Data

Display the first observations, structure and sample size:

```
head(ab_clean)
str(ab_clean)
nrow(ab_clean)
```

Task 4: Compute Point Estimators

```
summary_stats <- ab_clean %>%  
group_by(group) %>% summarize( total_users = n(), conversions = sum(converted),  
  conversion_rate = mean(converted) )  
summary_stats
```

Theory: Two-Sample Proportion Test

Suppose

$$\hat{p}_1$$

and

$$\hat{p}_2$$

are the sample conversion rates. Under

$$H_0 : p_1 = p_2$$

the pooled estimator is

$$\hat{p} = \frac{x_1 + x_2}{n_1 + n_2}.$$

The test statistic is

$$Z = \frac{\hat{p}_1 - \hat{p}_2}{\sqrt{\hat{p}(1 - \hat{p}) \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}}.$$

Task 5: Perform the Test in R

Extract the required quantities:

```
successes <- summary_stats$conversions  
trials <- summary_stats$total_users
```

Perform the test:

```
ab_test <- prop.test( successes , trials , alternative = "two.sided" , correct = FALSE )  
ab_test
```

Task 5: Perform the Test in R

Apply a two-sample independent t-test and construct the corresponding confidence interval for these data. Finally, compare the results of the t-test with those of the two-proportion z-test and interpret the results.

```
ab_t_test <- t.test(converted ~ group, data = ab_clean, alternative = "two.sided")  
ab_t_test
```

Questions?